

Van-Loc Nguyen

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EDUCATION

Univerisity of Science, VNUHCM

Master of Science in Computer Science, majoring in Artificial Intelligence.

Current GPA 4.00/4.00

Relevant Coursework: Mathematics for AI, Machine Learning, and Natural Language Processing.

Ho Chi Minh City, Vietnam

Dec 2024 - Present

Univerisity of Science, VNUHCM

Bachelor of Science in Computer Science, Honors Program

GPA: 3.88/4.00

Graduation Thesis Title: Fake News Analysis and Detection

Ho Chi Minh City, Vietnam

Oct 2020 - Oct 2024

Thesis Score: 10.00/10.00 (4.00/4.00)

EXPERIENCE

National Institute of Informatics (国立情報学研究所)

Research Intern

Chiyoda-ku, Tokyo, Japan

Oct 2025 - Present

Autonomous Inc

AI Engineer

Ho Chi Minh City, Vietnam

Mar 2025 - Aug 2025

- Build fascinating AI agents with OpenAI and Langchain for real-world applications.
- Turn existing MCP servers into AI agents.
- Optimize AI agents to perform stably on local environment (MacOS-centric) with small models (for example, Qwen-4B) by context engineering.

SELab, University of Science, VNUHCM

Research Assistant

Ho Chi Minh City, Vietnam

Jan 2025 - Present

- Participate in research in the multimedia field.
- Research Interests: Computer Vision, Deep Learning, Multimedia Verification, Generative AI Application.

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Backend Engineer

Ho Chi Minh City, Vietnam

Jun 2023 - Feb 2025

- Developed, optimized, and maintained the backend side for several projects, with a significant number of users in the US market, in Django and Django REST Framework, combined with PostgreSQL and Celery.
- Familiar with Version Control System, especially Git, be aware of Git flow and GitHub flow.
- Experienced with DevOps tools such as Kubernetes or Teleport.

Backend Engineer Intern

Mar 2023 - May 2023

- Be trained with essential skills for a back-end developer, especially for Python with Django, Docker usage, and PostgreSQL.
- Be aware to provide a clean codebase, with a strict follow of the PEP8.

PUBLICATIONS

Dinh-Khoi Vo, **Van-Loc Nguyen**, Minh-Triet Tran, and Trung-Nghia Le (2025). EVENT-Retriever: Event-Aware Multimodal Image Retrieval for Realistic Captions. In *Proceedings of the 33rd ACM International Conference on Multimedia (ACM MM'25)*. Accepted, published in October 2025.

Dinh-Khoi Vo, **Van-Loc Nguyen**, Minh-Triet Tran, and Trung-Nghia Le (2025). SAMURAI: Shape-Aware Multi-modal Retrieval for 3D Object Identification. *2025 International Conference on Multimedia Analysis and Pattern Recognition (MAPR)*. Accepted, published in August 2025.

Gia-Huy Vuong, **Van-Loc Nguyen**, Van-Son Ho, Tien-Thanh Nguyen Dang, Ngoc-Do Tran, Van-Tu Ninh, and Minh-Triet Tran (2025). NewsInsight2.0: An Enhanced Version Integrating Large Language Model-Based Query Optimization with Advanced Temporal Mechanisms. In: *Information and Communication Technology. SOICT 2024. Communications in Computer and Information Science, vol 2353*. Springer, Singapore, DOI: 10.1007/978-981-96-4291-5_28.

Thanh-Son Nguyen*, Viet-Tham Huynh*, **Van-Loc Nguyen***, and Minh-Triet Tran (2024). An Approach to Complex Visual Data Interpretation with Vision-Language Models. In *Proceedings of the Asian Conference on Computer Vision (ACCV) Workshops* (pp. 334-350), DOI: 10.1007/978-981-96-2641-0_23.

Minh-Triet Tran, Bao-Tin Nguyen, **Van-Loc Nguyen**, Thanh-Son Nguyen, Mai-Khiem Tran, and Trong-Le Do (2024). AI-Enhanced Photo Authenticity: A User-Focused Approach to Detecting and Analyzing Manipulated Images. *2024 International Conference on Multimedia Analysis and Pattern Recognition (MAPR)*, Da Nang, Vietnam, 2024, pp. 1-6, DOI: 10.1109/MAPR63514.2024.10661051.

Bao-Tin Nguyen*, **Van-Loc Nguyen***, Thanh-Son Nguyen, Duc-Tien Dang-Nguyen, Trong-Le Do, and Minh-Triet Tran. 2024. A Hybrid Approach for Cheapfake Detection Using Reputation Checking and End-To-End Network. In *Proceedings of the 1st Workshop on Security-Centric Strategies for Combating Information Disorder (SCID '24)*. Association for Computing Machinery, New York, NY, USA, Article 3, 1-12. DOI: 10.1145/3660512.3665521.

Van-Loc Nguyen*, Bao-Tin Nguyen*, Thanh-Son Nguyen, Duc-Tien Dang-Nguyen, and Minh-Triet Tran. 2024. A Unified Network for Detecting Out-Of-Context Information Using Generative Synthetic Data. In *Proceedings of the 2024 International Conference on Multimedia Retrieval (ICMR '24)*. Association for Computing Machinery, New York, NY, USA, 1300-1305. DOI: 10.1145/3652583.3657599.

PROJECTS

ArcanaAI: AI-Powered Tarot Reading Service

Apr 2025 - Aug 2025

An AI-powered web application for interactive tarot readings, supporting multiple decks, daily draws, and custom spreads.

Integrated **OpenAI GPT models** with **function calling** for personalized Tarot interpretations in real time.

Developed an interactive chat interface with journaling, history tracking, and multi-language readiness.

Built with FastAPI, Langchain, Next.js 15, and Docker.

Role: Main Developer

Deployment Link: <https://arcanaai.nguyenvanloc.com>

GitHub URL: <https://github.com/vanloc1808/arcana-ai>

SanghaGPT: Multilingual Buddhist Q&A Chatbot

July 2025

A multilingual chatbot for question answering on Buddhist texts, supporting both English and Vietnamese.

Integrated large language models (GPT-4o) via Model Context Protocol (MCP) for context-aware, high-accuracy responses.

Achieved the cosine similarity of 84.65% over 235 evaluation samples.

Engineered robust prompt strategies and pipelines for multilingual information retrieval and response synthesis.

Deployed with FastAPI and a web-based user interface.

Role: Main Developer

Deployment Link: <https://sanghagpt.nguyenvanloc.com>

GitHub URL: <https://github.com/hcmus-project-collection/sanghagpt>

GraphRAG: Knowledge-Graph Retrieval System

March 2025

A RAG system combining vector search and knowledge graph traversal with local LLMs.

Automated triplet extraction and graph construction from documents.

Developed a hybrid retrieval strategy (vector search and knowledge graph search) to improve accuracy and reduce hallucination.

Deployed with FastAPI and a minimal web interface.

Role: Main Developer

GitHub URL: <https://github.com/hcmus-project-collection/graphrag-llm-with-knowledge-base>

Fake News Analysis and Detection

Feb 2024 - Aug 2024

Undergraduate thesis at the University of Science, VNUHCM.

Propose the usage of an end-to-end network for the Visual Entailment task to solve both tasks of the cheapfakes detection challenge.

Introduce a cheapfakes data augmentation method with LLMs.

Implement a reputation checking module to check the image over the Internet to utilize social information.

Develop an open system to assist users in fake news analysis and detection with the help of an external forensic framework.

Role: Co-Main Developer.

GitHub URL: <https://github.com/nbtin/fakenews-detection-demo>

SKILLS & INTERESTS

Technical: Python, PyTorch, FastAPI, Django, C++.

Language: English, Vietnam.

Soft Skills: Team work, communication.

CERTIFICATES & AWARDS

Third Prize, The 26th Student Scientific Research Award - Euréka 2024 Ho Chi Minh City, Vietnam
Student Scientific Research Award –Euréka is a prestigious award for students interested in scientific research across the country, contributing to the promotion of the scientific research movement of university students. The Ho Chi Minh Youth Union of Ho Chi Minh City hosts the award in collaboration with the National University of Ho Chi Minh City.
Dec 2024

Third Prize, The 2024 “AI City Challenge” (AIC) of Ho Chi Minh City Ho Chi Minh City, Vietnam
The 2024 AI City Challenge is hosted by the Department of Information and Communications of Ho Chi Minh City, aiming to provide information retrieval solutions on Vietnamese news data.
Dec 2024

Certificate of Satisfactory Progress from President of Vietnam National University Ho Chi Minh City Ho Chi Minh City, Vietnam
This certificate is awarded by the President of Vietnam National University, Ho Chi Minh City, to students who have graduated with a high distinction degree.
Dec 2024

OTHER ACTIVITIES

Reviewer of the 1st Workshop on Ambiguous Object Analysis in Computer Vision (AOCV) at ACM Multimedia 2025 (ACM MM’25).